ADVANCED PROGRAMMING LANGUAGES

Spring 2026, CM 0632. MSc in Computer Science and Information Technology, Ca' Foscari University of Venice

Review - Class exercises

λ Calculus

1. α-Equivalence

Which of these expressions are α-equivalent (equal up to renaming)?

1. $\lambda x.y\ \lambda a.a\ x$
2. $\lambda x.z\ \lambda b.b\ x$
3. $\lambda a.y\ \lambda b.b\ a$

> **Solution**

- 1 and 3 are alpha equivalente
- 2 is not equivalent to either one

2. How many β-redexes do you see?

- (λx. x) ((λy. y) 5)
- (λf. f 1) (λx. (λy. y) x)
- (λx. x) ((λy. y) ((λz. z) 5))

> **Solution**

- There are two redexes in the first term, three redexes in the second and third terms.

3. Reduce each of the above by

- call by name

> **Solution**

- Redexes highlighted in red

$$\overline{(\lambda x.x)((\lambda y.y)5) \rightarrow (\lambda y.y)5}$$

$$\overline{(\lambda f.f\ 1)(\lambda x.(\lambda y.y)x) \rightarrow (\lambda x.(\lambda y.y)x)\ 1}$$

$$\frac{}{(\lambda x.x)((\lambda y.y)((\lambda z.z)5)) \to (\lambda y.y)((\lambda z.z)5)}$$

- call be value

> **Solution**

- Redexes highlighted in red

$$\frac{\dfrac{}{(\lambda y.y)5 \to 5}}{(\lambda x.x)((\lambda y.y)5) \to (\lambda x.x)5}$$

$$\frac{}{(\lambda f.f1)(\lambda x.(\lambda y.y)x) \to (\lambda x.(\lambda y.y)x)\,1}$$

$$\frac{\dfrac{}{(\lambda z.z)5 \to 5}}{(\lambda x.x)((\lambda y.y)((\lambda z.z)5)) \to (\lambda x.x)((\lambda y.y)5)}$$

Inductive definitions

Let the set Q be defined inductively as follows:

$$\frac{}{8 \in \mathsf{Q}}(\mathrm{R1}) \qquad \frac{}{5 \in \mathsf{Q}}(\mathrm{R2}) \qquad \frac{a \in \mathsf{Q} \quad b \in \mathsf{Q}}{a+b+1 \in \mathsf{Q}}(\mathrm{R3})$$

4. Give a derivation showing that $14 \in \mathsf{Q}$.

> **Solution**

$$\frac{\dfrac{}{8 \in \mathsf{Q}} \quad \dfrac{}{5 \in \mathsf{Q}}}{14 \in \mathsf{Q}}(14 = 8+5+1)$$

5. Give a derivation showing that $20 \in \mathsf{Q}$.

> **Solution**

$$\frac{\dfrac{\dfrac{}{8 \in \mathsf{Q}} \quad \dfrac{}{5 \in \mathsf{Q}}}{14 \in \mathsf{Q}} \qquad \dfrac{}{5 \in \mathsf{Q}}}{20 \in \mathsf{Q}}$$

6. State the principle of rule induction for Q.

- That is, if you want to prove that a property $P(a)$ holds for all $a \in \mathsf{Q}$, what must you show?

Solution

We must show:

- *Base clauses*: $P(8)$ and $P(5)$
- *Inductive clauses*: For any given a and b, $P(a) \wedge P(b) \implies P(a+b+1)$

7. Prove or disprove that $2 \in \mathsf{Q}$

- To prove it, give a derivation
- To disprove, find a property that holds of all elements of Q but not of 2.

Solution

- Clearly, $2 \notin \mathsf{Q}$
- To prove it, we can define $P(a) \triangleq a > 2$ and use rule induction to show that $P(a)$ holds forall $a \in \mathsf{Q}$

Typed Lambda Calculus

8. Type checking by hand. Construct a typing derivation for the following terms and determine their types (or report a type error):

- $\lambda x : \mathsf{Nat}.\ \mathsf{if}\ (x = 0)\ \mathsf{then\ true\ else\ false}$
- $\lambda f : (\mathsf{Nat} \to \mathsf{Bool}).\ \lambda x : \mathsf{Nat}.\ f\ (x+1)$
- $(\lambda x : \mathsf{Nat}.\ x)\ \mathsf{true}$

Solution

$$\dfrac{\dfrac{\dfrac{\dfrac{}{x : \mathsf{Nat} \vdash x : \mathsf{Nat}} \quad \dfrac{}{x : \mathsf{Nat} \vdash 0 : \mathsf{Nat}}}{x : \mathsf{Nat} \vdash x = 0 : \mathsf{Bool}} \quad \dfrac{}{x : \mathsf{Nat} \vdash \mathsf{true} : \mathsf{Bool}} \quad \dfrac{}{x : \mathsf{Nat} \vdash \mathsf{false} : \mathsf{Bool}}}{x : \mathsf{Nat} \vdash \mathsf{if}\ (x = 0)\ \mathsf{then\ true\ else\ false} : \mathsf{Bool}}}{\vdash \lambda x : \mathsf{Nat}.\ \mathsf{if}\ (x = 0)\ \mathsf{then\ true\ else\ false} : \mathsf{Nat} \to \mathsf{Bool}}$$

$$\dfrac{\dfrac{\dfrac{}{f : \mathsf{Nat} \to \mathsf{Bool}, x : \mathsf{Nat} \vdash f : \mathsf{Nat} \to \mathsf{Bool}} \quad \dfrac{\dfrac{}{f : \mathsf{Nat} \to \mathsf{Bool}, x : \mathsf{Nat} \vdash x : \mathsf{Nat}} \quad \dfrac{}{f : \mathsf{Nat} \to \mathsf{Bool}, x : \mathsf{Nat} \vdash 1 : \mathsf{Nat}}}{f : \mathsf{Nat} \to \mathsf{Bool}, x : \mathsf{Nat} \vdash x + 1 : \mathsf{Nat}}}{f : \mathsf{Nat} \to \mathsf{Bool}, x : \mathsf{Nat} \vdash f\ (x+1) : \mathsf{Bool}}}{f : \mathsf{Nat} \to \mathsf{Bool} \vdash \lambda \mathsf{x} : \mathsf{Nat}.\ \mathsf{f}\ (\mathsf{x}+1) : \mathsf{Nat} \to \mathsf{Bool}}}{\vdash \lambda f : (\mathsf{Nat} \to \mathsf{Bool}).\ \lambda x : \mathsf{Nat}.\ f\ (x+1) : (\mathsf{Nat} \to \mathsf{Bool}) \to \mathsf{Nat} \to \mathsf{Bool}}$$

$$\dfrac{\dfrac{\dfrac{}{x : \mathsf{Nat} \vdash x : \mathsf{Nat}}}{\vdash \lambda x : \mathsf{Nat}.\ x : \mathsf{Nat} \to \mathsf{Nat}} \quad \dfrac{}{\vdash \mathsf{true} : \mathsf{Bool}}}{(\vdash \lambda x : \mathsf{Nat}.\ x)\ \mathsf{true} : \mathsf{error}}$$